

Q1: What are the steps to transform from Analog to digital?

A1: 1- Sampling

2- Quantizing

3- Coding

Q2: What is the information?

A2: Information is any knowledge that comes to our attention. That could be voice, image, text and video

Information is processed Data

Q3: state 4 advantages and disadvantages for the analog and the digital signals?

A3:

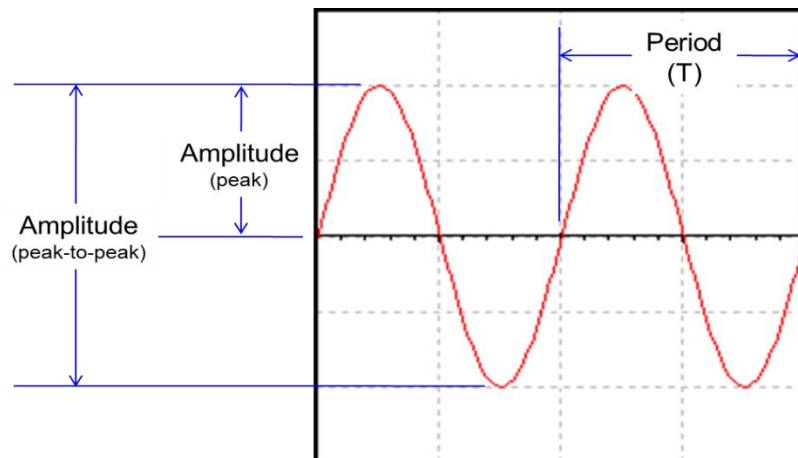
--Analog--	--Digital--
Continues	Discreet
Infinite value	Finite value

(Worth mentioning): 4 Advantages of Digital Technology:

1. (Most important one) Ability for noise removal in digital is easier than in analog
2. Capacity for error control
3. High speed
4. High level of security

Q4: In the shown graph, state the variables?

A4:



Q5: What is the frequency?

A5: A measure of the number of occurrences of the signal per second. (Hertz, Hz)

Q6: Matching definitions

Q7: What is the difference between the internet and the www? State 2.

A7: Web contains the information.

Internet transports information to and from users

(Worth mentioning): www is a part of the Internet

Q8: What are the steps to run the machine cycle?

A8: 1. Fetch

2. Decode

3. Execute

Q9: What is the internet?

A9: A global system of computers of thousands of privately and publicly owned computers and networks

Q10: What is the operating system? State his basic functions.

A10: Set of programs that coordinates:

Interactions of hardware components to each other

Interaction between application software and computer hardware

1. Starts the computer

2. Manages applications

3. Manages memory

4. Handles input and output device messages

5. Provides a user interface

Q11: What is paging?

A11: Transferring files between RAM and the hard disk

(Worth mentioning): it happens in the Virtual memory.

Q12: What is the file manager?

A12: utility software that organizes and manages data.

Q13: What is the back-up?

A13: copies data found on the hard disk to a backup device.

Q14: What is the Multitasking?

A14: permit more than one application to run at the same time.

Q15, 16, 17, and 18 are true and false.

15: speaker is an electronical device that is used to convert sound wave to digital wave.

False.

16: complex sound is made up of a number of pure sound component.

True.

17: error control is accomplished by playing extra bits, to transfer at the transmission.

True.

18: Haven't mentioned it.

Q19: how long it takes to download 600 s/of sound, speed 64, size 16 bit, sample 40 k/Hz.

number of sample = Sample * s/of sound = $40 * 600 = x$

Number of bits = $x * 16 = y$

$y / 64 = z$ (in second)

convert it to minute = $z / 60 = \text{Result}$

Q20: What does it mean network?

A20: A group of devices connected to each other to share information.